

EX NO: 10 CALCULATION OF PROCESSING GAIN AND GENERATION OF PN SEQUENCES FOR SPREAD SPECTRUM COMMUNICATION

Objectives

- i) To calculate the processing gain for various PN sequence lengths and tabulate the corresponding processing gain values.
- ii) To visualize and compare the processing gain obtained for different PN sequence lengths using MATLAB plots and analyze the effect of sequence length on spread spectrum performance.

CODE:

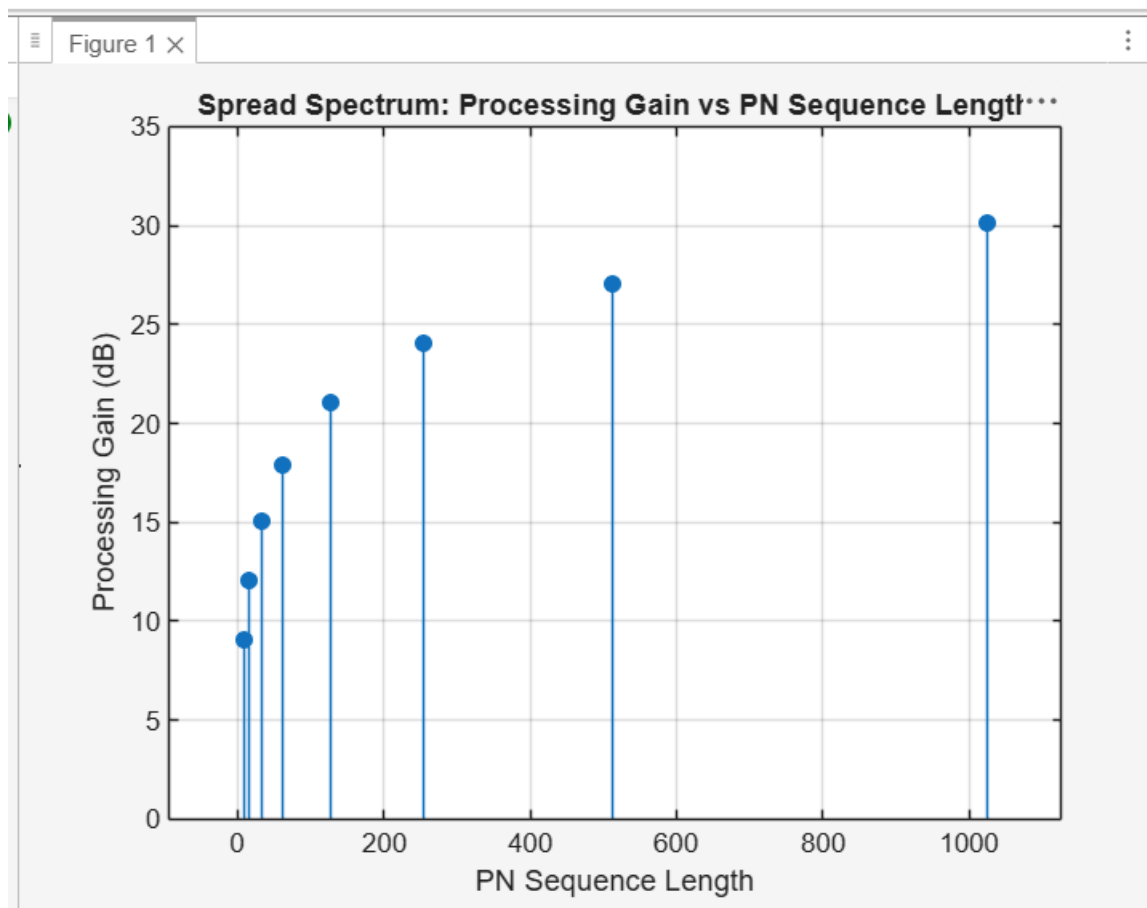
```
clc;
clear;
close all;
% PN sequence lengths
PN_length = [8 16 32 62 128 255 512 1024];
% Processing Gain (Linear)
PG_linear = PN_length;
% Processing Gain in dB
PG_dB = 10*log10(PG_linear);
% Display table
T = table(PN_length',PG_dB',...
    'VariableNames',{'PN_Length','Processing_Gain_dB'});
disp('Processing Gain Table');
disp(T);
% Plot
figure;
stem(PN_length,PG_dB,'filled');
grid on;
xlabel('PN Sequence Length');
ylabel('Processing Gain (dB)');
title('Spread Spectrum: Processing Gain vs PN Sequence Length');
```

Processing Gain Table

PN_Length Processing_Gain_dB

8	9.0309
16	12.041
32	15.051
62	17.924
128	21.072
255	24.065
512	27.093
1024	30.103

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Result:

The MATLAB plot successfully illustrated the relationship between PN sequence length and processing gain. Longer PN sequences produced higher processing gains, thereby improving interference rejection and communication reliability.